

Caroline “CC” LeBlanc

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EDUCATION

Master of Business Administration, May 2027

The University of Alabama, Manderson Graduate School of Business, Tuscaloosa, AL

GPA: 3.75/4.0 Focus Area of Study: Business Analytics

Bachelor of Science in Mechanical Engineering, May 2026

The University of Alabama, Lee J. Styslinger College of Engineering, Tuscaloosa, AL

GPA: 3.49/4.0 Honors College Fundamentals of Engineering (FE) Exam - Passed

SKILLS

MATLAB, Python (Beginner), Power BI (Beginner), Microsoft Excel, Motec i2, SolidWorks, Simulink, AutoCAD, Bosch WinDarab (Beginner), Cadmatic, WinGeo3, Adobe Creative Cloud, TIG Welding

WORK

EXPERIENCE

Engineering Intern, May 2026 – August 2026

BF Goodrich / Michelin, Tuscaloosa, AL

- Implemented a profilometer measurement system on treadline operations, bringing tread cut angle from 83° to 90° and raising process capability (Cp) from 1.02 to 1.12
- Designed new skiver blade components in SolidWorks to improve tread cut quality and reduce variability along treadline systems
- Conducting flow analysis of spray bath systems in tread manufacturing lines to identify process optimization opportunities
- Gained hands-on knowledge of tire manufacturing processes across carcass building and tread extrusion processes

Engineering Intern, May 2025 – August 2025

Heinen & Hopman, Spakenburg, The Netherlands

- Performed heat transmission calculations in company software to cross-check Excel-based hand calculations, proving the need for continuous software development
- Gained hands-on experience with AutoCAD to produce ventilation and HVAC system drawings for marine applications

PROJECT

EXPERIENCE

Vehicle Dynamics Senior Engineer, Suspension Assistant Lead, May 2024 – May 2026

Alabama FSAE (Crimson Racing), The University of Alabama, Tuscaloosa, AL

- Diagnosed a rear-axle oscillation through data analysis and retuned dampers to solve the issue, reducing lateral-g oscillation amplitudes by 56%, ultimately helping the team place 6th overall at FSAE Michigan in 2026
- Redesigned rubber bumpstops to be 3D-printed TPU parts. Tested 54 samples and reduced manufacturing variation by 83%
- Developed a Pacejka-based Magic Formula tire model in MATLAB to extract raw data and characterize lateral and longitudinal force behavior, cutting tire selection decision time by 50%
- Designed and deployed driver feedback setup sheets, standardizing quantitative data collection (Cooper-Harper scale, tire pressures, camber, toe) to enable setup comparison across test days

Communications Lead, May 2025 – present

Alabama FSAE (Crimson Racing), The University of Alabama, Tuscaloosa, AL

- Lead a subteam of 4, providing mentorship in Adobe software to ensure sustainable knowledge transfer among members
- Boosted engagement by 75% through outreach events, recruitment initiatives, and social media campaigns
- Designed and managed all digital and print communications, including social media content, website updates, and university outreach materials such as flyers and promotional graphics

EDAG Student Team Lead, August 2025- May 2026

STEM GBA Capstone Project, The University of Alabama, Tuscaloosa, AL

- Organized timelines and project scope objectives for a student team of four
- Coordinated meetings and communicated final products to German clients to deliver a manufacturing facilities guideline

INVOLVEMENT

Alabama FSAE (Crimson Racing)

Alabama Collegiate Recovery Community

Society of Automotive Engineers